

Crowded Carry: crash risk in FX and crypto carry trades

A complete study guide to the project: the economics behind it, every step of the process from raw data to results, worked numerical examples, the full results, and what to say when the professor asks.

00 Summary in one minute

A **carry trade** borrows in a low-yield asset and invests in a high-yield one. It pays a small premium most of the time and occasionally crashes. The leading explanation is that the premium pays you for **crash risk**, and that crashes are worst when the trade is **crowded**: when many speculators hold it and must all unwind at once.

We built two carry strategies with the same logic in two very different markets and asked whether crowding helps predict the crashes:

- **A · FX Crowded G10 carry.** Long the 2 highest-rate developed currencies, short the 2 lowest, monthly. Crowding is measured from CFTC futures positioning. When the trade is crowded, we cut exposure in half.
- **C · crypto Crypto funding carry.** On Binance perpetual futures, short the third of coins with the highest funding rate and long the third with the lowest, rebalanced weekly.

A CARRY, NET SHARPE

0.37

1999 to 2026. Max drawdown
−37% (2008).

A WITH CROWDING FILTER, OUT
OF SAMPLE

0.32

vs 0.27 without the filter (2011 to
2026).

C CRYPTO CARRY, NET SHARPE

0.14

−0.07 in 2020–23, 0.81 from
2024. Max drawdown −67%.

CORRELATION OF A AND C

0.11

A 50/50 mix at 10% vol has
Sharpe 0.51.

The one-sentence story. Carry in both markets behaves like a reward for crash risk; in FX, crowding is a modest but useful warning sign; in crypto, the strategy's crashes come from single coins (LUNA, FTT), and because FX and crypto carry crash at different times, they diversify each other.

01 The big idea

Why should carry make money at all?

Theory says it should not. **Uncovered interest parity (UIP)** says a high-rate currency should depreciate by exactly the interest differential, so borrowing yen at 0.1% to hold Australian dollars at 4% earns nothing on average. In the data, UIP fails: high-rate currencies do not depreciate enough on average, and often appreciate. This is the **forward premium puzzle**, and it is why carry earns money.

Then why doesn't everyone do it?

Because the profits come with **negative skewness**: many small gains and a few large, sudden losses. The saying is that carry goes "up the stairs and down the elevator". Losses cluster in bad times for everyone (2008, sudden volatility spikes), so the premium can be read as compensation for risk, not a free lunch.

Where does crowding come in?

Brunnermeier, Nagel and Pedersen (2008) show that carry crashes happen when speculators are heavily positioned in the trade and funding conditions tighten. If everyone is long AUD and short JPY, a shock forces them to sell AUD and buy JPY at the same time, and their own selling deepens the crash. So: **a crowded carry trade should have more crash risk**. That is the hypothesis we test in FX, using CFTC positioning as the crowding gauge.

Why crypto?

Crypto perpetual futures have their own interest rate: the **funding rate**. When many traders are leveraged long a coin, they pay a high funding rate to the shorts. So a high funding rate is both a carry signal (shorts get paid) and a crowding signal (lots of leveraged longs). Same economics, completely different investors and market structure. It lets us ask whether the carry and crash logic carries over.

An analogy

Carry is like selling insurance. You collect a premium every month and pay out rarely, but when you pay, it is large and comes at the worst time. Crowding is like many insurers having written the same policy: when disaster comes, they all try to offload risk at once.

The two markets side by side

FX yield = interest differential, crowding = CFTC positions.

Crypto yield = funding rate, and funding itself measures leveraged demand.

02 Concepts you need

Grouped from the foundations up. Each card gives the definition and how this project uses it.

Currency foundations

Spot rate S

Price of one unit of foreign currency in USD. We store all rates as USD per foreign unit, so S going up means the foreign currency appreciated.

FRED series like DEXUSAL are already USD per AUD; DEXJPUS (yen per USD) is inverted.

Interest differential

Foreign 3-month rate minus the US 3-month rate, $i_k - i_{US}$. This is the carry signal in Strategy A.

Data: OECD 3-month interbank rates via FRED (IR3TIB01...M156N).

Covered interest parity (CIP)

Forward = Spot $\times (1 + i_{US}) / (1 + i_K)$. It means buying a currency forward is the same as borrowing dollars and depositing in the foreign currency. That is why we can compute carry returns from spot and rates without forward data.

Uncovered interest parity (UIP)

Expected depreciation equals the interest differential. If it held, carry would earn zero. Its empirical failure is the forward premium puzzle.

Currency excess return rx

The return of borrowing USD and holding foreign currency for one month: foreign interest plus currency move, minus US interest.

Formula and worked example in section 5.

Dollar-neutral portfolio

Long and short legs have equal size, so the portfolio has no net bet on the dollar. It bets only on high-rate currencies beating low-rate ones.

Crowding and positioning

CFTC Commitments of Traders

Weekly US regulator report of positions in futures. For each currency future at the CME it splits traders into commercial (hedgers) and non-commercial (speculators).

Positions as of Tuesday, published Friday. We use them only after Friday.

Net speculative positioning

(Non-commercial long – non-commercial short) / open interest. Positive means speculators are net long the foreign currency.

z-score

(today's value – rolling mean) / rolling standard deviation over 36 months. It says how unusual today's positioning is compared with recent history.

Portfolio crowding

Average z of the long leg minus average z of the short leg. High means speculators are unusually long what we are long and short what we are short. Above 1 we call the trade crowded.

Crypto mechanics

Perpetual future (perp)

A futures contract with no expiry. To keep its price close to spot, longs and shorts exchange a periodic payment called funding.

Funding rate

Paid every 8 hours on Binance. If positive, longs pay shorts; if negative, shorts pay longs. Typical value 0.01% per 8 hours, about 11% a year.

Our data: average funding about 12–15% a year on major coins, positive on 81–88% of days.

Funding as carry

Being short a high-funding coin earns funding, like being long a high-rate currency earns interest. Being long a negative-funding coin also earns funding.

Survivorship bias

Testing only on coins that still exist today makes results look better than they were. We keep LUNA and FTT, which collapsed in 2022.

Statistics and backtesting

Sharpe ratio

Annualised mean excess return divided by annualised volatility. Monthly: $\text{mean} \times 12 / (\text{sd} \times \sqrt{12})$. Daily crypto: $\times 365$.

Maximum drawdown

Largest fall from a previous peak in cumulative value. Captures crash risk that volatility misses.

Skewness

Asymmetry of returns. Negative skew means occasional large losses. FX carry is negatively skewed (−0.49).

Turnover and costs

Turnover = sum of absolute weight changes.
Cost = turnover $\times$ cost per unit: 3 bp for FX, 5 bp for crypto.

In-sample / out-of-sample

Results on data before the split (A: to 2010, C: to 2023) versus data after it. Out-of-sample is the fairer test.

Look-ahead bias

Using information not yet known at decision time. We avoid it by forming weights at the end of period t and earning period $t+1$'s return, and by lagging CFTC data to its release date.

Newey-West t-statistic

A t-statistic that corrects standard errors for autocorrelation and heteroskedasticity in time-series regressions. We use 3 lags (monthly) and 7 lags (daily).

Volatility targeting

Scale a strategy so its predicted volatility equals a target (10% a year), using only past volatility. Makes two strategies comparable before combining them.

03 Research questions and hypotheses

These were written down before running any backtest. Writing them first is what makes the test honest.

#	Question	Hypothesis (H1)
RQ1 A	Does G10 carry earn a positive return after costs, and does it match the academic carry factor?	Positive premium; high correlation with the Lustig–Roussanov–Verdelhan (LRV) HML factor.
RQ2 A	Does CFTC crowding predict weaker or more crash-prone carry, and does cutting exposure when crowded help out of sample?	After crowded months, returns are lower and more negatively skewed; the filter raises the out-of-sample Sharpe.
RQ3 A	When does FX carry lose money?	Negative beta to volatility shocks; positive exposure to the dollar factor.
RQ4 C	Does a short-high-funding, long-low-funding portfolio earn a premium after costs, and where does it come from?	Positive premium, mainly from funding collected.
RQ5 C	What drives crypto carry losses, and does a simple risk control help?	Little BTC exposure; losses come from single coins when crowded positions unwind.
RQ6	Are FX and crypto carry correlated? Does combining them diversify?	Low correlation; the mix has lower volatility and drawdown.

04 The pipeline, start to end

Seven numbered scripts in `code/`, run in order by `run_all.py`. The notebooks in `notebooks/01–05` walk through the same steps by importing the same functions, so their numbers match.

01

Download data for A. FX spot (FRED H.10), 3-month rates (OECD via FRED), CFTC Commitments of Traders zips 1986–2026. Cleans everything to month-end, converts to USD per foreign unit, matches CFTC contract names (including the pre-2000 "INTERNATIONAL MONETARY MARKET" listings).

→ data/clean/fx_spot_monthly_usd_per_fcu.csv, rates_3m_monthly.csv, cftc_net_spec_weekly.csv

02

Backtest A. Excess returns, rank by rate differential, long 2 / short 2, crowding z-scores, filter, costs, performance tables, crowded-vs-not test, robustness grid, figures.

→ output/strategyA/

03

Download data for C. Binance public archive (data.binance.vision): daily candles and 8-hourly funding for 12 USDT perpetuals, 2020–2026. Funding summed to daily; a coin is tradable only on days with price and positive volume.

→ data/clean/ crypto prices, volume, funding

04

Backtest C. 7-day funding signal, weekly sort into thirds, daily P&L split into price and funding, costs, performance, decomposition, robustness grid (3/7/30 days × daily/weekly), worst weeks.

→ output/strategyC/

05

Risk analysis. Compare A to the LRV factor (course file CurrencyPortfolios.xls), regress A on dollar and volatility factors, C beta to BTC, P&L by coin, pre-specified risk control for C, combined 50/50 portfolio at 10% vol.

→ output/risk/

06

Interactive page. Builds the results website from the output tables.

→ web/index.html

07

Slides. Builds the 16-slide deck with charts from output/.

→ slides/presentation.pptx (+ PDF)

How the project actually went

1. **Oct 1.** Chose strategies A and C from the course's list. Set up the GitHub repo and a Python 3.12 environment.
2. **Data trouble.** FRED blocked scripted downloads from the network, so the FRED CSVs were saved by hand. CFTC files downloaded fine.
3. **Bug found and fixed.** CFTC positions first started only in 2000 because the code matched only "CHICAGO MERCANTILE". Before August 2000 the same contracts are listed under "INTERNATIONAL MONETARY MARKET". Matching both names extended positioning back to 1986.
4. **Crypto universe.** Cut to 12 coins, deliberately keeping LUNA and FTT to avoid survivorship bias, and USDC only as a sanity check.
5. **Risk analysis, website, slides, notebooks, README, AI log.** All pushed to `main` by Oct 2.

05 Strategy A, step by step FX

Universe: JPY, EUR, GBP, CHF, CAD, AUD, NZD, each against USD. **Frequency:** monthly.

Sample: 1999-05 to 2026-08 (limited by the start of JPY and CHF rate data and the euro).

Step 1. Compute each currency's monthly excess return

$$rx(k, t) = (1 + i_{k,t-1} / 1200) \times S_t / S_{t-1} - (1 + i_{US,t-1} / 1200)$$

i in % per year, so $/1200$ turns it into a monthly decimal. S in USD per foreign

WORKED EXAMPLE

- a AUD rate 4.0%, US rate 1.0%. AUD/USD moves from 0.7000 to 0.7035 (+0.5%).
- b $rx = (1 + 0.04/12) \times 1.005 - (1 + 0.01/12) = 1.003333 \times 1.005 - 1.000833 = 1.008350 - 1.000833 = \mathbf{+0.75\%}$.
- c If AUD instead falls 1%: $1.003333 \times 0.99 - 1.000833 = \mathbf{-0.75\%}$.
- d Break-even: AUD can fall about 0.25% a month (3% a year) before carry loses. UIP says it should fall exactly that much on average; in the data it doesn't, which is the carry premium.

Step 2. Rank and form the portfolio

At each month-end, rank the 7 currencies by $i_k - i_{US}$. Go long the top 2 and short the bottom 2. Each long has weight +0.5, each short -0.5 (dollar-neutral).

WORKED EXAMPLE (ILLUSTRATIVE RATES)

Currency	3m rate	Diff vs USD (3.5%)	Rank	Weight
NZD	5.5%	+2.0	1	+0.5
AUD	4.3%	+0.8	2	+0.5
GBP	4.0%	+0.5	3	0
CAD	3.0%	-0.5	4	0
EUR	2.0%	-1.5	5	0
CHF	0.5%	-3.0	6	-0.5
JPY	0.1%	-3.4	7	-0.5

Next month's portfolio return = $0.5 \times (r_{NZD} + r_{AUD}) - 0.5 \times (r_{CHF} + r_{JPY})$.

Step 3. Measure crowding with CFTC data

1. For each currency future: net speculative position = (non-commercial long – short) / open interest.
2. Shift each Tuesday report by 3 days so it is used only after its Friday release; take the last value in each month.
3. z-score each currency over a rolling 36-month window (at least 24 months needed).
4. Portfolio crowding = $\sum w_k \times z_k$ = average z of the longs – average z of the shorts.

WORKED EXAMPLE

- a z-scores: NZD +1.8, AUD +1.4 (speculators unusually long), CHF -0.6, JPY -1.2 (unusually short).
- b Crowding = $0.5(1.8) + 0.5(1.4) - 0.5(-0.6) - 0.5(-1.2) = 0.9 + 0.7 + 0.3 + 0.6 = 2.5$.
- c $2.5 > 1$, so the trade is crowded: all weights are multiplied by 0.5 for next month.

Step 4. Timing and costs

Weights decided at the end of month t earn month $t+1$'s return. Turnover = $\sum |w_t - w_{t-1}|$, and the cost is turnover $\times$ 3 bp.

Example: GBP replaces AUD in the long leg. AUD goes $0.5 \rightarrow 0$ and GBP $0 \rightarrow 0.5$, so turnover = 1.0 and the cost is 0.03% that month. Halving exposure when crowded also creates turnover, which is why the filtered strategy's turnover is higher (2.0 $\times$ vs 1.2 $\times$ a year).

Pre-specified parameters

Parameter	Value	Why
Currencies per leg	2	Standard for a 7-currency universe; keeps legs diversified.
z-score window	36 months (min 24)	Long enough to define "normal" positioning, short enough to adapt.
Crowding threshold	1.0	One standard deviation; robustness grid also tests 0.5 and 1.5.
Exposure when crowded	50%	Reduce, don't exit; grid also tests 0%.
CFTC release lag	3 days	Tuesday data published Friday.
Cost	3 bp per unit	Typical G10 forward bid-ask.
Out-of-sample start	2011-01	Leaves 2008 in-sample and 15+ years for the test.

06 Strategy C, step by step crypto

Universe: 12 Binance USDT-margined perpetuals: BTC, ETH, BNB, XRP, ADA, DOGE, SOL, LTC, LINK, AVAX, plus LUNA and FTT, which collapsed in 2022. **Frequency:** weekly rebalance, daily P&L. **Sample:** 2020-02 to 2026-08.

Step 1. Build the signal

Sum each day's three 8-hour funding payments into daily funding. Signal = trailing 7-day average of daily funding, using data up to the previous day.

Step 2. Sort into thirds every Sunday

Among coins that are tradable that day (price and positive volume), with at least 6 coins available: $k = \text{number of coins} / 3$. Short the k coins with the highest funding (weight $-0.5/k$ each), long the k with the lowest ($+0.5/k$ each). Each leg is 50% of capital, so the portfolio is dollar-neutral with gross exposure 1.

Step 3. Daily P&L = price + funding

$$\begin{aligned} \text{price P\&L} &= \sum w_{k,t-1} \times \text{return}_{k,t} \\ \text{funding P\&L} &= - \sum w_{k,t-1} \times \text{funding}_{k,t} \quad (\text{a short, } w < 0, \text{ receives positive}) \\ \text{net} &= \text{price} + \text{funding} - \text{turnover} \times 5 \text{ bp} \end{aligned}$$

WORKED EXAMPLE: ONE DAY SHORT DOGE

- a 11 coins tradable, so $k = 3$ and each short has weight $-0.5/3 = -16.7\%$.
- b DOGE funding today: 0.04% (about 15% a year). Funding P&L = $-(-0.167) \times 0.04\% = \mathbf{+0.0067\%}$.
- c DOGE price rises 2% today. Price P&L = $-0.167 \times 2\% = \mathbf{-0.33\%}$.
- d One ordinary price move wipes out about 50 days of funding. This is the key fact behind Strategy C's results: the funding income is real but price moves dominate.

Step 4. Handling dead coins

A coin is tradable only on days with a price and positive volume. That removes LUNA after 2022-05-12 and FTT after 2022-11-14 (Binance's archive shows a frozen FTT price after that). A coin that stops trading between rebalances is closed out. Only 5 days missing from Binance's archive are forward-filled.

Pre-specified parameters

Parameter	Value
Funding lookback	7 days
Rebalance	Weekly (Sunday close)
Leg size	Top and bottom third
Minimum coins to trade	
Cost	5 bp per unit of turnover
Out-of-sample start	2024-01-01
Risk control (tested separately) Cap 1/6 of a leg per coin; no long in a coin with 7-day funding below −50% a year	

07 Keeping the test honest

Parameters fixed first

Every window, threshold and cost was written down before running backtests and never changed afterwards.

Out-of-sample split

A: 1999–2010 vs 2011–2026. C: 2020–2023 vs 2024–2026.

No look-ahead

Weights at t earn $t+1$. CFTC used only after release. Crypto signals use funding up to the previous day.

Costs included

All headline numbers are net of costs; gross is also reported.

No survivorship bias

LUNA and FTT stay in the crypto universe.

Robustness reported, not tuned

Grids show alternative parameters; the headline stays the pre-specified case even when another looks better.

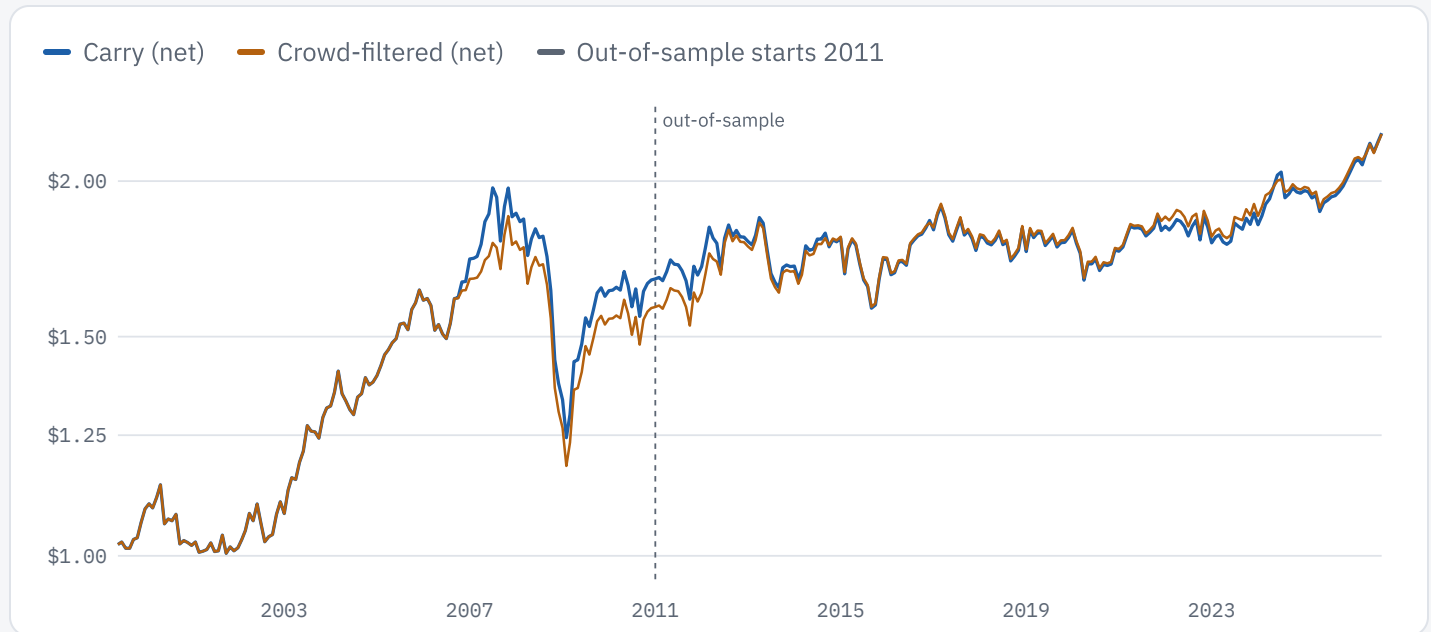
External validation

Strategy A is checked against the published LRV developed-country carry factor.

One risk control, fixed in advance

Reported next to the base case whatever the outcome, which turned out to be negative.

08 Results: Strategy A FX



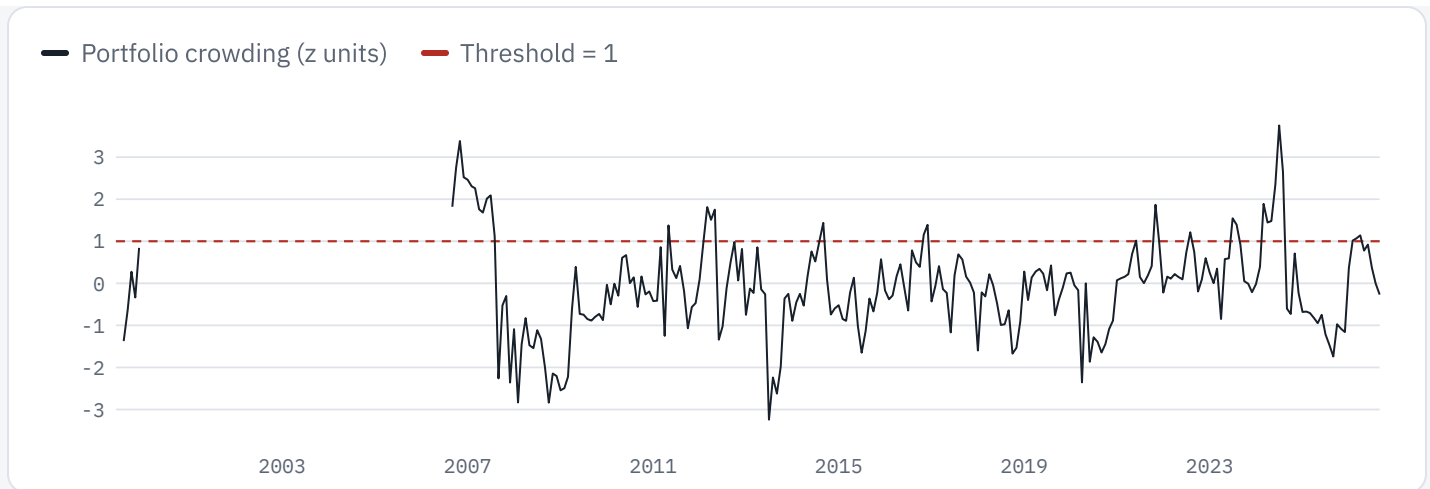
Growth of \$1, log scale, 1999-05 to 2026-08. \$1 became about \$2.19 in both versions. The big drop is 2008.

Performance (net of costs)

	Period	Ann. mean	Ann. vol	Sharpe	Max DD	Skew	Turn
Carry	Full 1999–2026	3.27%	8.91%	0.37	−37.0%	−0.49	
Carry	In-sample 1999–2010	4.96%	10.55%	0.47	−37.0%	−0.71	
Carry	Out-of-sample 2011–2026	2.01%	7.46%	0.27	−15.5%	−0.10	
Crowd-filtered	Full	3.23%	8.55%	0.38	−37.0%	−0.45	
Crowd-filtered	In-sample	4.47%	10.12%	0.44	−37.0%	−0.69	
Crowd-filtered	Out-of-sample	2.31%	7.16%	0.32	−14.4%	−0.02	

How to read it. Carry earns about 3% a year with Sharpe 0.37: modest but positive. Returns are lower after 2010 (low rates everywhere shrank differentials). The filter helps a little out of sample (0.27 → 0.32) and reduces vol and skew, but cannot avoid the 2008 drawdown.

Crowding signal over time



Crowding above 1 happened in 33 months. Gaps are months where a z-score was unavailable (for example, no NZD futures contract from 2000 to 2003).

Does crowding predict bad months? (RQ2)

	Months	Next-month mean	Next-month vol	Skewness
Not crowded	295	0.30%	2.56%	-0.46
Crowded	33	0.02%	2.67%	-0.71

After crowded months, carry earns almost nothing and is more negatively skewed, exactly the direction the hypothesis predicts. But 33 months is a small sample, so the difference is not statistically strong.

Robustness grid (net Sharpe)

Threshold	Exposure when crowded	Sharpe full	Sharpe out-of-sample	Max DD
0.5	0%	0.37	0.31	-37.0%
0.5	50%	0.38	0.30	-37.0%
1.0	0%	0.38	0.37	-37.0%
1.0 (pre-specified)	50%	0.38	0.32	-37.0%
1.5	0%	0.35	0.40	-37.0%
1.5	50%	0.36	0.33	-37.0%

Every variant beats the unfiltered out-of-sample Sharpe of 0.27, so the improvement isn't an accident of one parameter choice.

Validation against the academic factor (RQ1)

Over 1999-05 to 2021-05 (264 months), Strategy A has a **0.82 correlation** with the LRV developed-country HML carry portfolio from the course file. Strategy A: 3.22% a year, Sharpe 0.34. LRV HML: 2.34%, Sharpe 0.23. Our implementation reproduces the known carry factor.

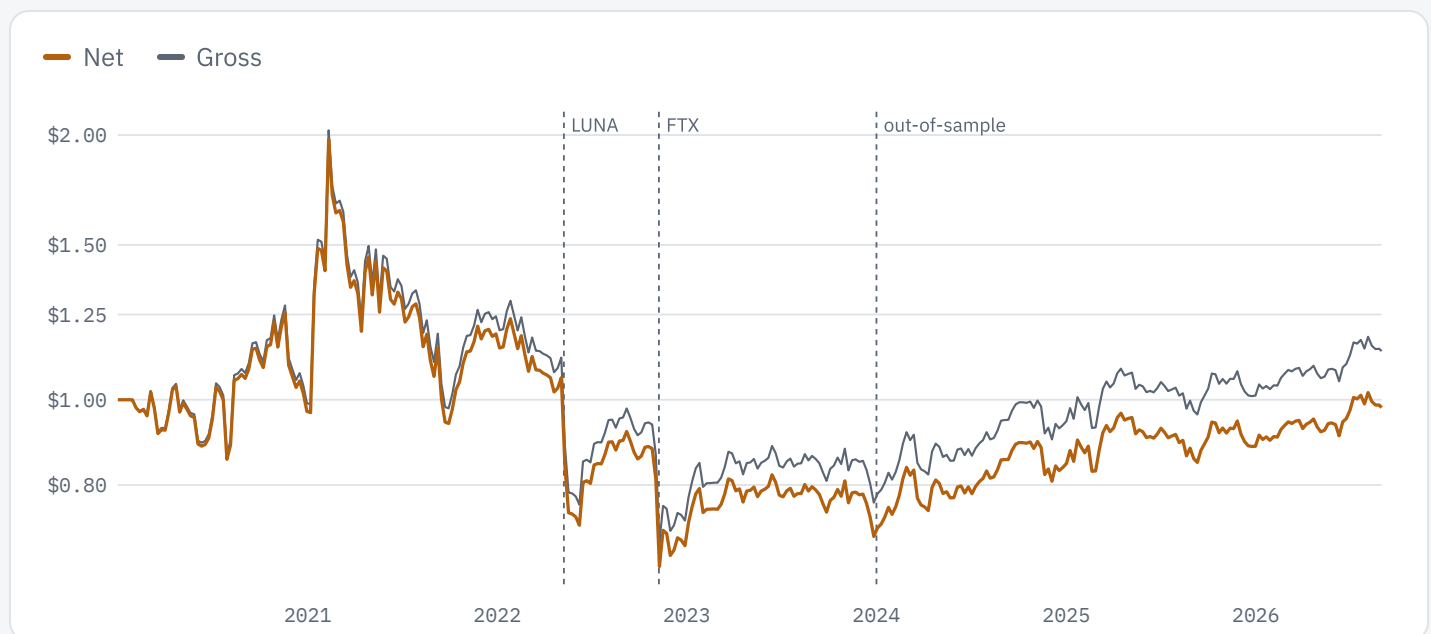
When does it lose? (RQ3)

Model	Dollar factor RX	Volatility innovation	Monthly alpha	R ²
RX only	0.42 (t = 4.4)	–	0.26% (t = 1.6)	0.12
Vol only	–	–1.93 (t = –4.3)	0.26% (t = 1.6)	0.10
Both	0.35 (t = 4.1)	–1.55 (t = –3.7)	0.25% (t = 1.7)	0.18

How to read it. Carry loses when global equity volatility jumps (strongly significant negative beta). After controlling for the dollar and volatility, the alpha is not significant ($t \approx 1.7$). That is the textbook result: carry is compensation for risk, not free money.

Worst months: 2008-10 (–12.0%), 2007-08 (–7.8%), 2000-05 (–7.0%), 2009-01 (–6.8%), 2008-03 (–6.5%), 2015-01 (–6.4%, the Swiss franc de-peg). July 2008 to March 2009: –20.5% in total, as JPY and CHF (the funding currencies) rallied.

09 Results: Strategy C crypto

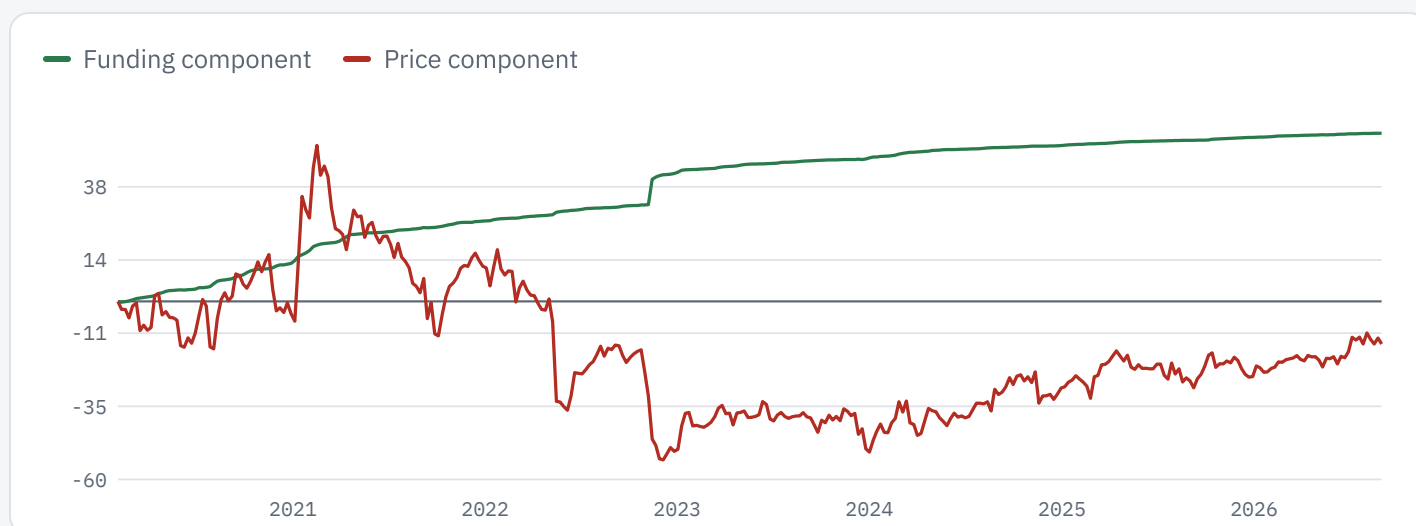


Growth of \$1, log scale. Marked: LUNA (May 2022), FTX (Nov 2022), out-of-sample start (2024).

	Period	Ann. mean	Ann. vol	Sharpe	Max DD	Worst week	Turnover
Gross	Full 2020–2026	6.38%	29.87%	0.21	−66.0%	−27.2%	45
Net	Full	4.13%	29.87%	0.14	−67.3%	−27.3%	45
Net	In-sample 2020–2023	−2.52%	36.05%	−0.07	−67.3%	−27.3%	45
Net	Out-of-sample 2024–2026	13.87%	17.21%	0.81	−13.3%	−9.2%	44

Puzzle to understand: the net mean is +4.1% a year, yet \$1 ends at about \$0.98. Why? With 30% volatility, the compounded (geometric) return is roughly the arithmetic mean minus half the variance: $4.1\% - 0.5 \times 0.30^2 \approx 4.1\% - 4.5\% \approx -0.4\%$ a year. This is volatility drag.

Where the return comes from (RQ4)



Cumulative contribution, percentage points.

% per year	Funding	Price	Costs	Net
Full	+8.55	−2.17	−2.25	+4.13
In-sample	+12.26	−12.51	−2.27	−2.52
Out-of-sample	+3.10	+12.99	−2.22	+13.87

Funding income is steady and real. In 2020–2023, the shorted coins (crowded longs like DOGE and XRP) kept rallying and wiped it out. Since 2024 the gain came mostly from price moves, not funding, so the strong out-of-sample number is not clean evidence of a stable funding premium.

Robustness: lookback × rebalancing (net Sharpe)

Lookback	Rebalance	Sharpe full	Sharpe out-of-sample	Max DD	Turnover
3 days	Daily	0.68	0.41	−40.5%	160×
3 days	Weekly	0.49	1.29	−61.2%	52×
7 days	Daily	0.56	0.92	−59.7%	97×
7 days (pre-specified)	Weekly	0.14	0.81	−67.3%	45×
30 days	Daily	0.28	0.83	−73.3%	33×
30 days	Weekly	0.07	0.42	−82.2%	19×

Shorter, faster signals do better even after costs, which suggests funding information decays within days. We still report the pre-specified case as the main result; switching after seeing the grid would be data snooping.

What drives the losses? (RQ5)

- **No market exposure:** beta to daily BTC returns is -0.02 ($t = -0.75$). The strategy is market-neutral.
- **Losses are coin-specific.** Total P&L by coin (percentage points): LUNA -87 , XRP -58 , DOGE -44 , FTT -13 ; SOL $+102$, BNB $+51$, AVAX $+36$, ADA $+32$, LINK $+27$.
- **The LUNA week** (ending 2022-05-15) lost 27.3%. The strategy was *long* LUNA: traders were shorting it, so its funding turned negative, and the rule puts lowest-funding coins in the long leg.
- Other worst weeks: 2021-09-12 (-12.3%), 2022-11-13 FTX (-11.6%), 2020-11-29 (-11.0%), 2021-08-15 (-10.0%).

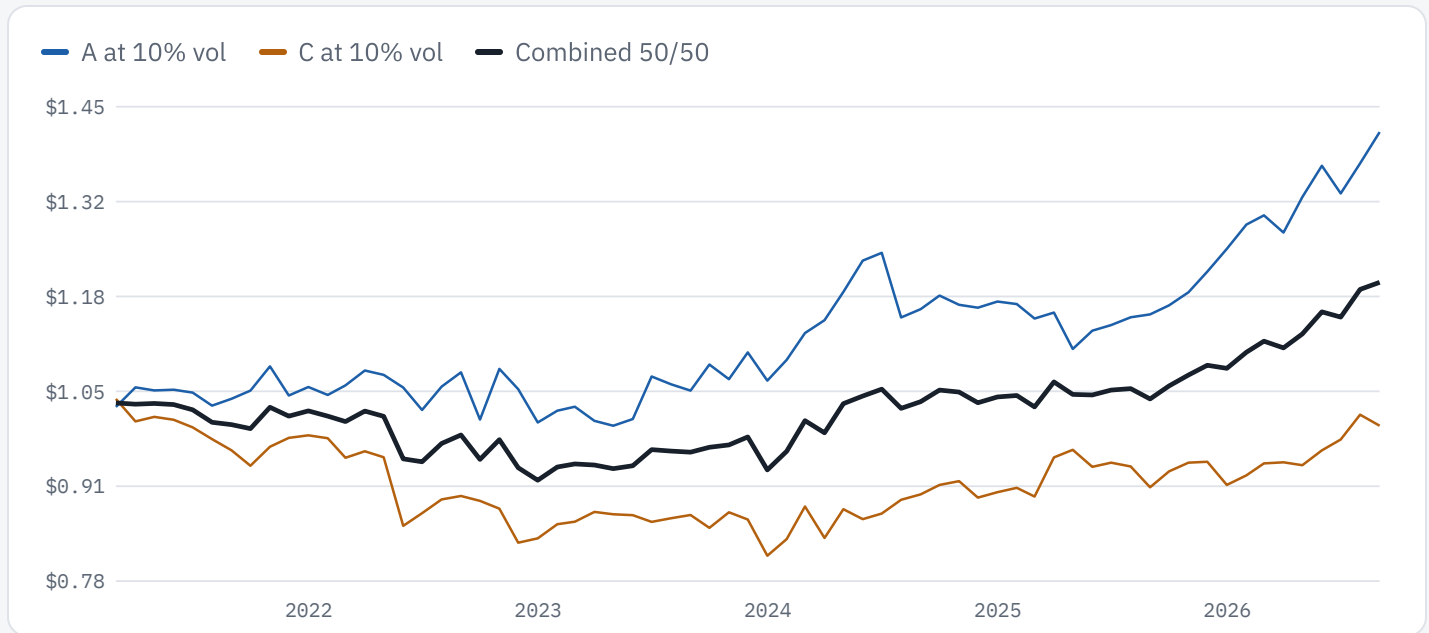
Mirror image of FX. FX carry crashes when crowded *longs* unwind. Crypto funding carry ends up long whatever traders short hardest, so it crashes when a heavily shorted coin collapses for real.

The pre-specified risk control

Net	Sharpe full	Sharpe out-of-sample	Max DD	Worst week	Avg gross exposure
Base	0.14	0.81	-67.3%	-27.3%	1.00
Risk-controlled	-0.16	0.74	-43.3%	-18.9%	0.49

It halved the LUNA and FTX losses, but mostly because the 1/6 cap left about half the capital unused. LUNA's funding crossed -50% only mid-week (10 May 2022), after the Sunday rebalance, and the filter blocked BNB and SOL, the two biggest winners. Lesson: in crypto, deeply negative funding usually signals crowded *shorts*, whose forced covering pushes prices up, not distress.

10 Combined portfolio (RQ6)



Growth of \$1, 2021-02 to 2026-08 (67 months). Each strategy scaled to 10% volatility using only past volatility (36 months for A, 12 for C).

	Ann. mean	Ann. vol	Sharpe	Max DD	Worst month
A (10% vol)	6.70%	9.43%	0.71	-11.0%	-7.4%
C (10% vol)	0.44%	9.28%	0.05	-21.4%	-10.2%
Combined 50/50	3.57%	6.96%	0.51	-10.6%	-5.9%

Correlation 0.11. The mix has lower volatility and a smaller worst month than either part. It doesn't beat A alone on Sharpe because C was weak during these years, but the

diversification itself is clear.

11 What we conclude

Hypothesis	Verdict	Evidence
A1: G10 carry earns a premium	Supported	Net Sharpe 0.37; 0.82 correlation with LRV HML.
A2: Crowding predicts weaker carry; filter helps out of sample	Leaning yes	0.02% vs 0.30% next-month mean; OOS Sharpe 0.27 → 0.32 across the grid. Only 33 crowded months; did not avoid 2008.
A3: Carry loses in volatility spikes	Supported	Vol beta -1.55 , $t = -3.7$; alpha insignificant.
C1: Crypto funding carry earns a premium, mainly from funding	Mixed	Funding $+8.6\%$ a year, but net Sharpe only 0.14 and price moves dominate; IS -0.07 vs OOS 0.81.
C2: No BTC exposure; crashes are coin-specific	Supported	BTC beta ≈ 0 ; LUNA -87 pts, worst week -27% .
Risk control reduces losses without destroying returns	Not supported	Sharpe $0.14 \rightarrow -0.16$; missed LUNA timing.
FX and crypto carry diversify	Supported	Correlation 0.11; combined vol 7.0% vs $9.3\text{--}9.4\%$.

Implications. In both markets, carry looks like pay for bearing crash risk. In FX, positioning-based crowding is a cheap, modestly useful overlay. In crypto, funding measures leveraged demand, but sorting on it alone exposes you to single-coin collapses. A better crypto design would react faster than weekly, treat extreme negative funding as crowded shorts rather than a buy signal, and size positions by coin risk.

12 Limitations

- **Sample start.** Pre-euro DEXGEUS is missing, JPY rates start 2002 and CHF rates 1999, so Strategy A effectively starts in 1999-05.
- **Small crypto sample:** about 6.5 years, 12 coins, 3–4 coins per leg, so single events dominate.
- **Crypto IS/OOS reversal** (-0.07 vs 0.81). The out-of-sample period is short and driven by price, not funding.
- **Signal-horizon dependence:** faster signals look better, but we keep the pre-specified one.
- **LRV data end in 2021-05**, so the volatility regressions cover 1999–2021 only.

- **Simple costs:** no slippage under stress, borrow constraints, funding caps or exchange counterparty risk (FTX shows this is real).
- **Only 33 crowded months**, so the crowding effect is imprecise.

13 Questions to expect

Click to reveal a suggested answer. Try answering out loud first.

– Why can you compute carry returns without forward rates?

Covered interest parity: a forward position equals borrowing USD and depositing in the foreign currency. So the forward return is the foreign interest plus the spot change minus the US interest, which is our rx formula. CIP deviations exist (especially after 2008) but are small relative to monthly carry returns.

– Why use CFTC data rather than something else for crowding?

It is the only long, public, weekly measure of speculative positions in currencies, going back to 1986. It directly measures who holds the trade, which is the mechanism in Brunnermeier, Nagel and Pedersen. Its limit: it covers only exchange futures, a small part of the FX market, so it is a proxy.

– Why z-score positioning instead of using raw positions?

Normal positioning differs a lot by currency and changes over time (for example, speculators are structurally short JPY). The 36-month z-score asks "is this unusually crowded compared with recent history?", which is comparable across currencies.

– The filter didn't avoid 2008. Doesn't that kill your hypothesis?

It shows crowding is not the only trigger. 2008 began from a non-crowded reading; the shock came from funding markets and volatility, which our regressions capture (vol beta -1.55). Crowding still predicts weaker next-month returns (0.02% vs 0.30%) and improves out-of-sample Sharpe in every variant of the grid. So it is a useful overlay, not a crash detector.

- **How do you know there's no look-ahead bias?**

Weights formed at end of t earn $t+1$'s return (shift by one period in code). CFTC Tuesday positions are used only after the Friday release (3-day lag). Crypto signals use funding through the previous day. Rolling z-scores use only past data.

- **Crypto out-of-sample Sharpe is 0.81. Isn't that great?**

It's encouraging but we don't over-read it. It covers only about 2.7 years, and the decomposition shows most of the gain came from price moves (+13%), not funding (+3%). In-sample it was negative. We treat it as weak evidence, not a stable premium.

- **Why keep LUNA and FTT?**

To avoid survivorship bias. Dropping coins that died would remove exactly the crashes we want to study. LUNA alone cost 87 percentage points of P&L.

- **Why was the strategy long LUNA when it crashed?**

As LUNA fell, traders shorted the perp, pushing funding negative. The rule longs the lowest-funding coins (they pay you to be long). So the strategy bought what speculators were shorting hardest, right as it collapsed. It's the mirror image of FX carry's crash.

- **Why not switch to the 3-day daily version since it does better?**

Choosing the best variant after seeing results is data snooping and would overstate expected performance. We fixed parameters in advance and report the grid for transparency. The pattern (faster is better) is a finding for future work, not a reason to change the headline.

- **Why does the combined portfolio have a lower Sharpe than A alone?**

C had Sharpe 0.05 over 2021–2026, so mixing it in dilutes return. But volatility falls from about 9.4% to 7.0% and the worst month improves from -7.4% to -5.9% , because the correlation is only 0.11. With a better crypto leg, the diversification would raise the Sharpe.

– **How did you validate your FX implementation?**

Correlation 0.82 with the LRV developed-country HML factor over 264 overlapping months, similar mean and Sharpe. Differences come from our smaller universe (7 currencies), 2-per-leg sort, and different rate data.

– **What would you do with more time?**

Add pre-euro DEXGEUS to extend the FX sample; use VIX past 2021 for the volatility test; for crypto, use faster signals, treat extreme negative funding as a crowded-short warning, add coin-level risk sizing, and expand the coin universe.

14 Self-test cards

Active recall: read the question, answer in your head, then reveal. Shuffle to change the order.

1. What is a carry trade?

Borrow in a low-yield asset and invest in a high-yield one, earning the difference unless prices move against you.

2. What does UIP predict, and does it hold?

High-rate currencies depreciate by the interest differential, so carry earns zero. It fails empirically (forward premium puzzle), which is why carry earns money.

3. Write the monthly currency excess return formula.

$$rx = (1 + i_{k,t-1}/1200) \times S_t/S_{t-1} - (1 + i_{US,t-1}/1200)$$
, with S in USD per foreign unit.

4. Why can we use spot and interest rates instead of forwards?

Covered interest parity makes a forward position equivalent to borrowing USD and depositing in the foreign currency.

5. How is Strategy A formed?

Each month-end, rank 7 G10 currencies by rate differential vs USD; long top 2 (+0.5 each), short bottom 2 (−0.5 each).

6. How is portfolio crowding defined?

Average 36-month z-score of CFTC net speculative positioning in the long leg minus that of the short leg.

7. What happens when crowding > 1?

All Strategy A positions are scaled to 50% for the next month.

8. Why lag CFTC data by 3 days?

Positions are as of Tuesday but published Friday; using them earlier would be look-ahead bias.

9. What bug extended CFTC data back to 1986?

Pre-Aug-2000 contracts are listed under INTERNATIONAL MONETARY MARKET, not CHICAGO MERCANTILE; matching both names fixed it.

10. Strategy A full-sample net Sharpe and max drawdown?

0.37 and -37% (2008).

11. Out-of-sample Sharpe with and without the crowding filter?

0.32 with the filter vs 0.27 without (2011–2026).

12. Next-month carry after crowded vs non-crowded months?

0.02% vs 0.30%, with more negative skew (-0.71 vs -0.46). Only 33 crowded months.

13. Correlation of Strategy A with LRV HML?

0.82 over 1999-05 to 2021-05.

14. What is Strategy A's beta to volatility innovations (both-factor model)?

-1.55, $t = -3.7$. Carry loses when volatility spikes; alpha becomes insignificant.

15. What is a funding rate?

Periodic payment between longs and shorts on a perpetual future, every 8 hours on Binance. Positive: longs pay shorts.

16. How is Strategy C formed?

Each Sunday, rank tradable coins by 7-day average funding; short the top third, long the bottom third, each leg 50% of capital.

17. Strategy C net Sharpe: full, in-sample, out-of-sample?

0.14 full, -0.07 in 2020–23, 0.81 from 2024.

18. Where did Strategy C's return come from?

Funding +8.55% a year, price -2.17%, costs -2.25%, net +4.13%. Price dominated each sub-period.

19. Why did Strategy C hold LUNA long during its collapse?

Traders shorted LUNA, so funding went negative; the rule longs the lowest-funding coins.

20. Why does \$1 in Strategy C end near \$0.98 despite +4.1% mean?

Volatility drag: geometric $\approx$ arithmetic - $\sigma^2/2 = 4.1\% - 4.5\% \approx -0.4\%$ a year.

21. Did the crypto risk control work?

No. Full Sharpe fell to -0.16; it mostly halved exposure, missed LUNA timing and blocked BNB and SOL.

22. Correlation between A and C, and combined Sharpe?

0.11; combined 50/50 at 10% vol has Sharpe 0.51 with 7.0% vol.

23. Name three ways the backtest avoids bias.

Pre-specified parameters; $t \rightarrow t+1$ timing and release lags; keeping LUNA and FTT; costs included; out-of-sample split.

24. What's the mirror-image point between FX and crypto carry crashes?

FX carry crashes when crowded longs unwind; crypto funding carry crashes by being long the coins traders short hardest.

15 Code map and how to run it

File	What it does
code/01_download_data_strategyA.py	FX, rates, CFTC $\rightarrow$ data/clean/
code/02_backtest_strategyA.py	excess_returns, carry_weights, crowding_signal, backtest, perf $\rightarrow$ output/strategyA/
code/03_download_data_strategyC.py	Binance prices, volume, funding $\rightarrow$ data/clean/
code/04_backtest_strategyC.py	carry_weights (sort into thirds), backtest (price + funding – costs) $\rightarrow$ output/strategyC/
code/05_risk_analysis.py	LRV check, regressions, BTC beta, coin P&L, risk control, combined $\rightarrow$ output/risk/
code/06_build_web.py	Interactive results page $\rightarrow$ web/index.html
code/07_build_slides.py	Deck $\rightarrow$ slides/presentation.pptx
notebooks/01–05	Step-by-step versions of 01–05 for study

```
conda activate carry
pip install -r requirements.txt
SKIP_DOWNLOAD=1 python run_all.py      # rebuild everything from the committed c
python run_all.py                      # full pipeline including downloads
```

Repository: github.com/rezazamani2329/currency-crypto-carry.

All numbers come from the files in the repository's `output/` folder (main branch, results built 2026-10-02). Worked examples use illustrative numbers and are labelled as such.